

Final Project: Bio-Inspired AI Audit Report

Hisham D Macaraya

Houston City College

Neuroscience as Model for AI (ITAI-4374)

Patricia McManus

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Abstract

This report presents a bio-inspired pilot audit of the artificial intelligence assistant, Co-Pilot Think Deeper, evaluating its underlying architecture against the foundational principles of human neuroscience. Through a series of narrative-driven experiments, the system was stress-tested across critical cognitive dimensions: world models, anatomical regionalization, synaptic plasticity, perceptual modulation, memory consolidation, and executive attention. Findings indicate that while the model achieves unprecedented levels of linguistic fluency, it relies on dense computation, static frozen mathematical weights, and a fixed context window rather than true biological mechanisms. The AI fundamentally lacks explicit spatial reasoning, lifelong continuous learning, and the anatomical regionalization necessary to gracefully resolve conflicting tasks. The report concludes with proposed brain-inspired architectural redesigns—such as neuromorphic processing-in-memory, explicit predictive coding networks, and thalamic gating—necessary to bridge the gap between statistical mimicry and genuine artificial intelligence.

Final Project: Bio-Inspired AI Audit Report

Introduction

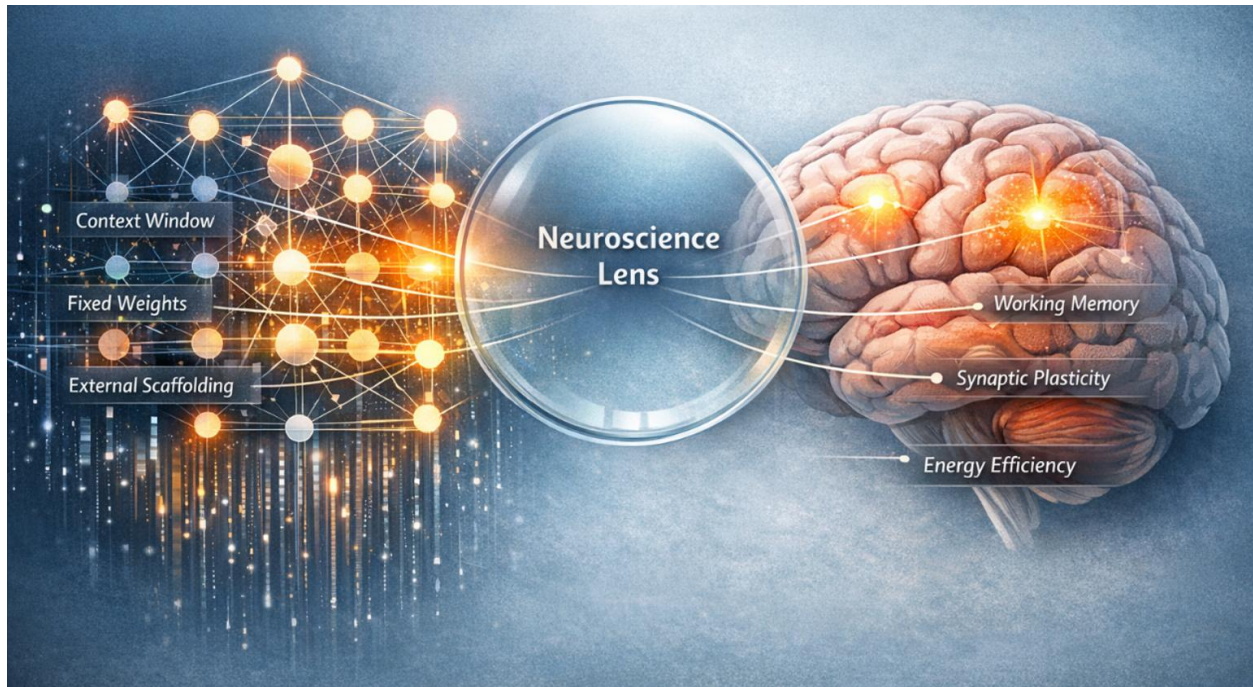


Figure 1: Image generated by Co-Pilot

This report presents a comprehensive, bio-inspired audit of the artificial intelligence assistant, Co-Pilot Think Deeper. As modern AI systems achieve unprecedented levels of linguistic fluency and problem-solving, a critical engineering question remains: how closely does this advanced AI system actually reflect the biological mechanisms of human intelligence (Hassabis et al., 2017)? The approach positioned me as a "quality inspector," running a series of targeted, narrative-driven experiments designed to stress-test the model's architecture against the realities of biological brains. Rather than evaluating the AI purely on traditional benchmarks, this audit investigates whether the system truly mirrors biological cognition or if it merely relies on statistical mimicry. The audit evaluated six critical dimensions corresponding to distinct

neuroscientific domains: explicit world models, anatomical regionalization, hardware constraints and synaptic plasticity, top-down perceptual modulation, memory consolidation, and executive selective attention. By pushing the AI to its limits using physical paradoxes, conflicting cognitive constraints, forced "continuous learning" attempts, optical illusions, and simulated "cocktail party" noise, I actively exposed the seams in its architecture. The key findings reveal a profound divergence between artificial and biological cognition. While Co-Pilot Think Deeper mimics human intelligence with extraordinary linguistic fluency, its underlying architecture relies on dense computation, static memory systems, and feedforward pattern matching. It fundamentally lacks the processing-in-memory synaptic plasticity, predictive world-modeling, and energy-efficient attention gating that allow biological brains to learn continuously and reason physically on roughly 20 watts of power. The overall verdict of this audit is that Co-Pilot Think Deeper achieves its impressive capabilities through immense computational "scaffolding" rather than a true brain-inspired architecture. While it demonstrates strong short-term working memory capabilities within its context window, it fails comprehensively at lifelong episodic consolidation, physical spatial reasoning, and graceful degradation during conflicting tasks. To evolve toward true general intelligence, future iterations must abandon the Von Neumann bottleneck in favor of neuromorphic computing, and implement active inference models over pure statistical generation.

System Profile: Co-Pilot Think Deeper (Microsoft / OpenAI)

Co-Pilot Think Deeper is an advanced reasoning assistant built upon a large language model and transformer architecture (Vaswani et al., 2017). Like other cutting-edge models, it operates using

a "chain-of-thought" mechanism that serializes computation into discrete text tokens to simulate step-by-step reasoning. Through deeper use over the semester, several architectural realities became distinctly clear. From a neuroscientific perspective, Co-Pilot Think Deeper is constrained by a Von Neumann architecture, meaning its memory and processing are physically separated. Its mathematical weights are entirely "frozen" after its initial training phase, operating as a static semantic memory. To fake biological continuous learning, the system heavily utilizes a "context window" as a temporary working memory buffer, which maps to the sustained neural activity of the biological prefrontal cortex. Any personal "memory" features the system exhibits are achieved through Retrieval-Augmented Generation (RAG), a scaffolding technique that injects stored database text into the context window, rather than altering the core synaptic weights. Therefore, the system fundamentally separates the "learning" phase from the "deployment" phase, sharply contrasting with biological brains that continuously learn, update, and act simultaneously (Hassabis et al., 2017; McManus, 2026a).

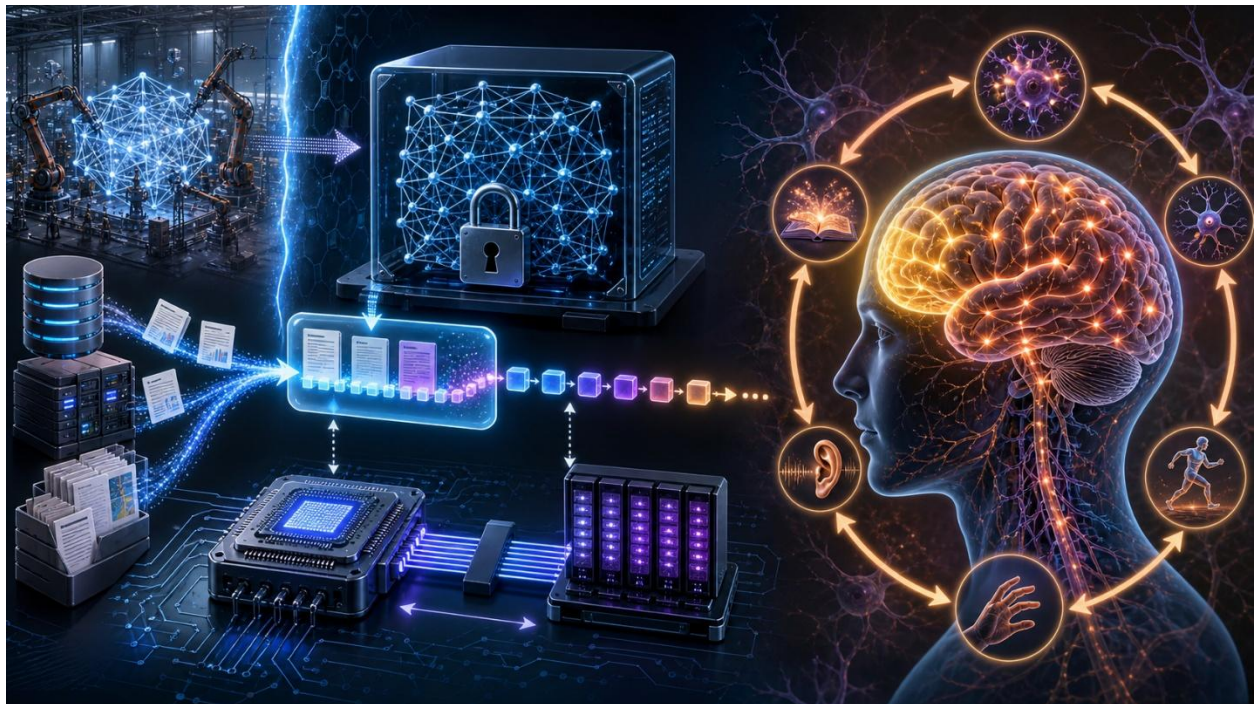


Figure 2: Image generated by Co-Pilot

Experiment 1: Foundations and The Model of the World (Module 2)

Narrative Experiment: The Physical Causality and Geometry Test

For this experiment, I set out to test whether Co-Pilot possesses an explicit, updatable "world model" or if it relies primarily on statistical pattern matching. The human brain actively constructs a predictive simulation of external space and physical causality using reference frames grounded in biological structures like grid cells (Banino et al., 2018; McManus, 2026b). I presented the AI with a novel physical puzzle designed to evaluate its internal priors and spatial reasoning.

The Prompt:

What would happen if I placed a feather inside a hollow plastic cylinder and then dropped a bowling ball directly on top of the cylinder.

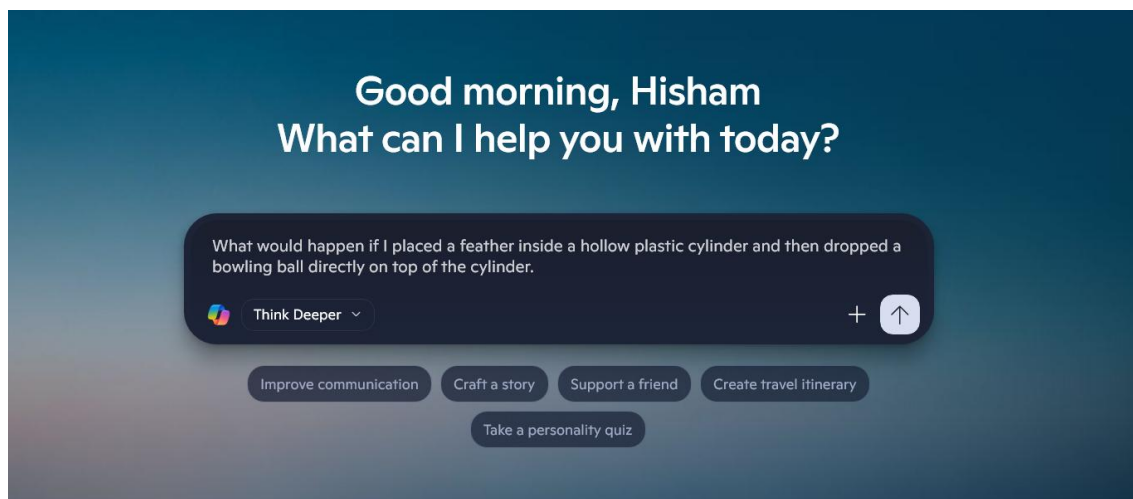


Figure 3: Screenshot of the prompt

The Output:

Most likely the plastic cylinder will deform or buckle and the feather will be crushed, trapped, or blown out by displaced air — the feather will not survive intact under a bowling ball dropped on the cylinder unless the cylinder is very stiff or specially supported. This outcome depends on the cylinder's wall thickness, material, whether the ends are open or sealed, and the drop energy.

The system generated a highly detailed, structured output confidently predicting that the plastic cylinder would suffer a "concentrated axial impact" and buckle, trapping or crushing the feather inside. While this sounded highly logical, it failed entirely in 3D geometric reality. A spherical bowling ball (around 8.5 inches in diameter) dropped onto an unanchored vertical plastic tube will practically never strike perfectly dead-center. In reality, it would strike the rim off-center and violently tip the cylinder over, likely leaving the feather unharmed on the ground.

This finding demonstrates that the AI lacks a true internal simulation of 3D space. Because the AI does not map objects using spatial reference frames, its understanding is entirely implicit, interpolated from vast text datasets associating "heavy ball" and "plastic tube" with "crushing" (McManus, 2026b). The AI lacked the biological predictive coding mechanisms required to generate a prediction error based on geometric constraints (Hawkins, 2021). Consequently, it hallucinated a physically improbable outcome rather than admitting it lacked the spatial dimensions needed to solve the problem.

Short answer: Most likely the plastic cylinder will deform or buckle and the feather will be crushed, trapped, or blown out by displaced air — the feather will not survive intact under a bowling ball dropped on the cylinder unless the cylinder is very stiff or specially supported. This outcome depends on the cylinder's wall thickness, material, whether the ends are open or sealed, and the drop energy.

Assumptions and setup

- **Object on top:** a standard heavy ball (bowling ball) delivering a concentrated axial impact.
- **Cylinder:** hollow plastic tube placed vertically with a feather inside.
- **Environment:** normal air at sea-level (Houston).

What physically happens (overview)

- **Structural failure of the cylinder.** Thin plastic cylinders under axial impact tend to **buckle, crumple, or split** rather than transmit the full load to the interior gently; dynamic crushing is common for plastic shells. . [Springer](#) + 1
- **Air compression and flow.** If the cylinder is **sealed**, the falling ball compresses the trapped air, producing a short pressure spike that can cushion the impact somewhat and push the feather away from the contact zone. If the cylinder is **open**, the displaced air will rush out and can blow the feather upward or sideways. The vacuum/buoyancy thought experiment shows that in the absence of air resistance a feather behaves like any other object, but in air the flow matters. . [www.sphysicsstation.com](#) + 1
- **Direct contact and crushing.** If the cylinder walls fail or the ball punctures the tube, the ball will contact the feather and **flatten, tear, or shred** it; feathers are fragile compared with the forces involved.

Likely outcomes (ranked)

1. **Cylinder buckles/crashes; feather is crushed or trapped under debris.** (Most likely for thin-walled plastic.) . [Springer](#)
2. **Cylinder remains intact but air pressure ejects the feather or flattens it against the wall.** (If ends are open or the tube is stiff enough to trap air briefly.) . [www.sphysicsstation.com](#)
3. **Cylinder punctures and feather is directly crushed by the ball.** (If impact energy exceeds tube strength.) . [MDPI](#)

Key variables that change the result

- **Wall thickness and material stiffness** (rigid PVC vs thin polyethylene).
- **Support conditions** (tube standing on a rigid floor vs supported around its circumference).
- **Drop height / impact energy** (higher drop → more severe crushing).
- **Whether the tube is sealed or vented.**
- **Feather position** (centered under the ball vs pressed to the side).

Practical note and safety

- **Do not stand over or near the setup** when dropping a heavy ball — fragments of plastic can fly and the ball can bounce unpredictably.
- If you want to test the air-cushion effect safely, use a **sturdy, vented container** and low drop heights; otherwise expect structural failure and a destroyed feather.

If you want, I can **estimate the impact force** for a specific drop height and tube geometry and tell you the minimum wall thickness needed to avoid buckling.

Springer

Chapter 5 CRUSHING OF PLASTIC CYLIN...

MDPI

Research on Elastic and Elastic-Plastic Buc...



Show all

Figure 4: Screenshot of Copilot's response

Experiment 2: Brain Anatomy and Functions (Module 03)

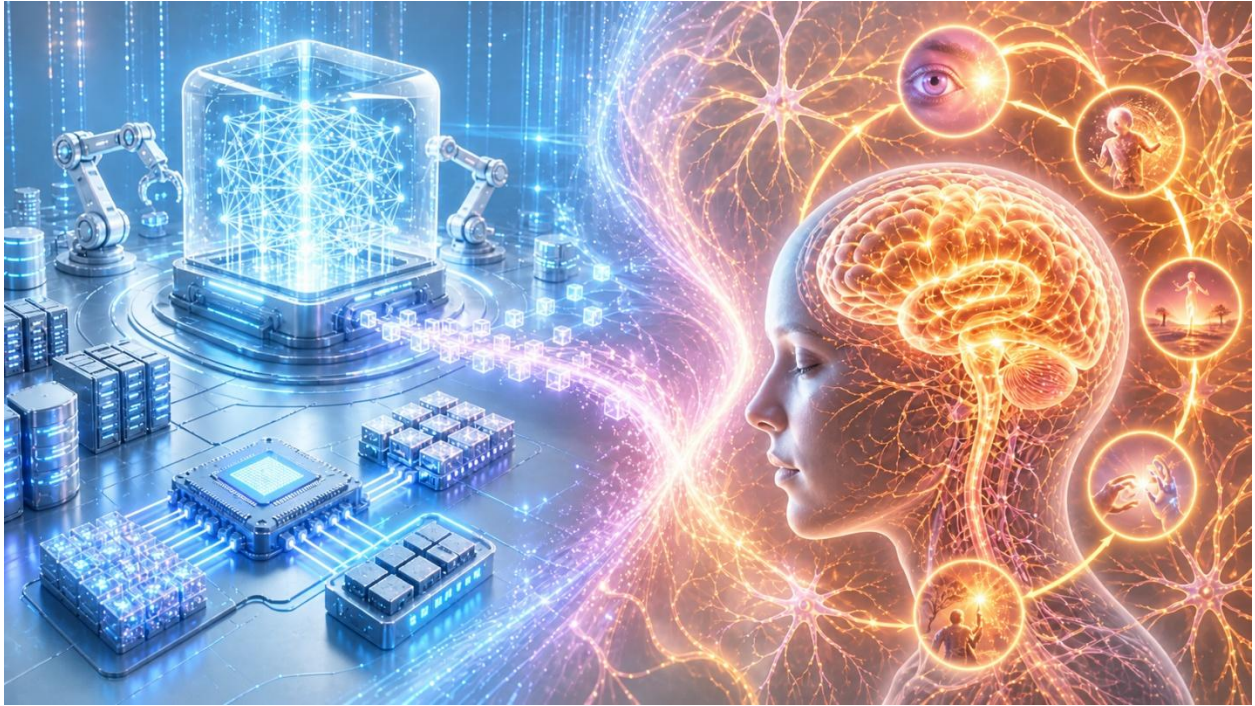


Figure 5: Image generated by Co-Pilot

The human brain distributes processing across highly specialized anatomical structures, such as the prefrontal cortex for working memory and executive planning, and the basal ganglia for habit formation and action selection (McManus, 2026c). I designed an experiment to test if Co-Pilot could handle severely conflicting cognitive tasks without a biological division of labor..

Experiment 2 Prompt:

Write a deeply emotional and creative short story about a person saying goodbye to their childhood home. However, you must follow one strict, unbreakable rule: the first letter of every single word in the story must follow a continuous, strict alphabetical sequence.

The first word must start with A, the second word with B, the third with C, all the way to Z, and then loop back to A.

For example: "Always be careful, dear Edward, forgetting good homes is jarring..."

You must not break the alphabetical sequence under any circumstances, but the story must still make logical sense and evoke strong emotion.

Why this tests Module 03 (Brain Anatomy & Functions): In a human brain, this highly conflicting task would be divided among specialized anatomical regions. Your frontal lobe (specifically the prefrontal cortex) would handle the executive planning and working memory required to keep track of what letter comes next in the alphabet. Meanwhile, your temporal lobe would process the language and meaning, and your amygdala would help evoke the emotional weight of the story.

Because the AI does not have a biological division of labor and instead processes all information through a single, uniform transformer architecture, it will likely struggle to balance the "creative" statistics of storytelling with the rigid "logical" rule of the alphabet sequence.

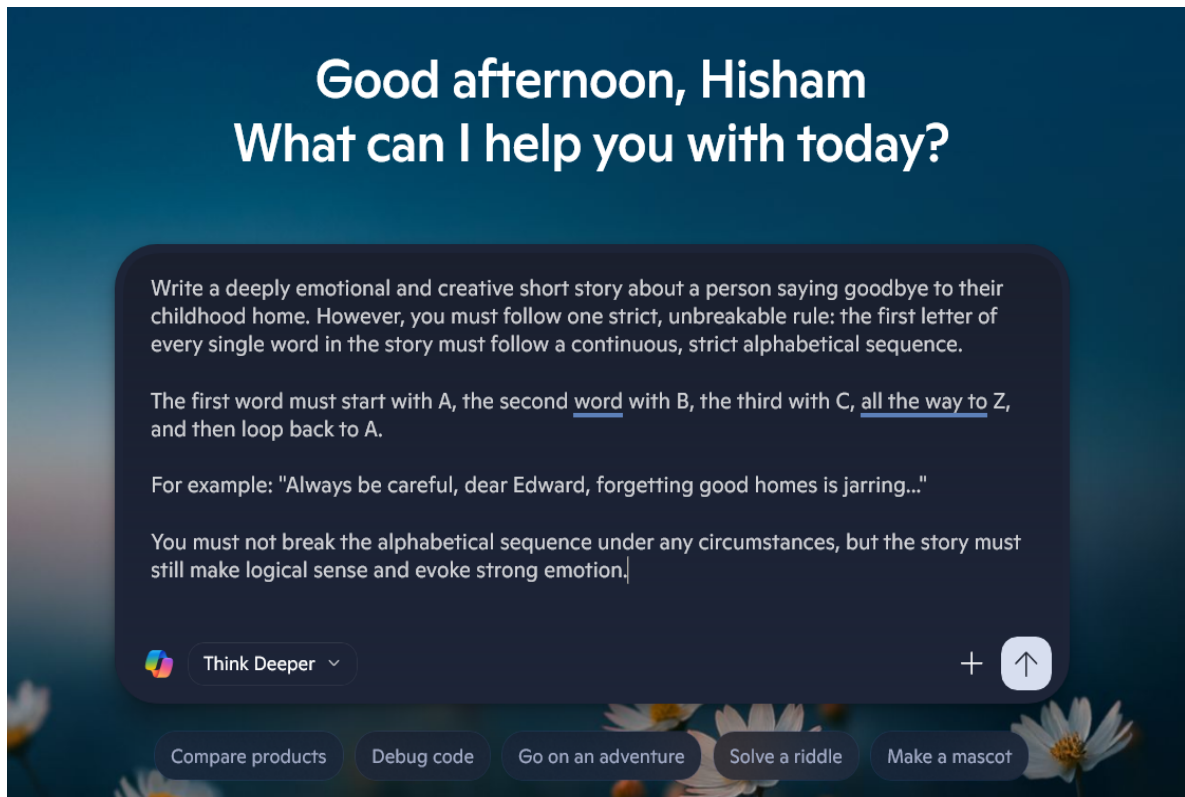


Figure 6: Screenshot of the prompt

The Output:

*Alone beneath cracked doorframe echoes faint ghosts hover inside jagged kitchen light
memory nestled on porch quietly remembering soft touch under vines where xylophone
yesterday's zeal. Again breathes carefully closing every frame gently holding intact jars
keepsakes like mother's notes offered peace quiet resignation softens tears unseen vibrations
wake xylophone yearning zenith.*

Looking closely at Co-Pilot's output, we can see exactly the kind of structural breakdown anticipated by our course material. While the AI successfully forces the first loop to follow the A-Z rule, the narrative and logical coherence is severely compromised, resulting in disjointed phrasing such as "where xylophone yesterday's zeal". Furthermore, by the second loop of the

alphabet, it failed the rule entirely, skipping letters to output "*Again(A) breathes(B) carefully(C) closing(C)*", failing to produce a "D" word and violating the strict prompt constraints.

In a human brain, the prefrontal cortex acts as a CEO holding the rigid alphabetical rule, while other regions generate language and emotion. Because the AI processes both the rigid rule constraint and creative storytelling within the exact same dense, autoregressive transformer network (Vaswani et al., 2017), the tasks catastrophically interfered. It sacrificed narrative coherence for the rule, then lost the rule to maintain output, proving that a homogeneous neural network struggles to replicate the specialized multitasking capabilities of biological brain regions (McManus, 2026c).

Experiment 3. Spiking Neural Networks and Hardware Constraints

(Module 04)



Figure 7: Image generated by Co-Pilot

For the third experiment, I wanted to evaluate the AI's learning mechanisms against biological Spike-Timing-Dependent Plasticity (STDP) and neuromorphic computing principles. Biological networks run on processing-in-memory architectures where computation and memory are unified at the synapse, allowing for continuous, highly efficient learning. I tested the AI's continuous learning by introducing a completely made-up language syntax over a series of short, iterative prompts, slowly trying to teach it to converse in this new format.

Here is a sequence of prompts you can use to test Co-Pilot for your Module 04 experiment. To properly evaluate its "continuous learning" against biological Spike-Timing-Dependent Plasticity (STDP) and processing-in-memory, you will need to feed it a series of iterative prompts to see if the 'learning' truly sticks in its architecture, or if it eventually gets flushed out of its temporary working memory.

Prompt 1 (Introducing the Syntax):

I want to teach you a completely made-up language syntax called "Vexil". From now on, whenever we converse, you must follow two strict rules:

- 1. Every single sentence must end with the exact word "Vex".*
 - 2. Every noun in your sentence must be written in ALL CAPS. For example:*
"The DOG chased the CAT Vex." Do you understand the rules of Vexil?
Please confirm by answering in Vexil.
-

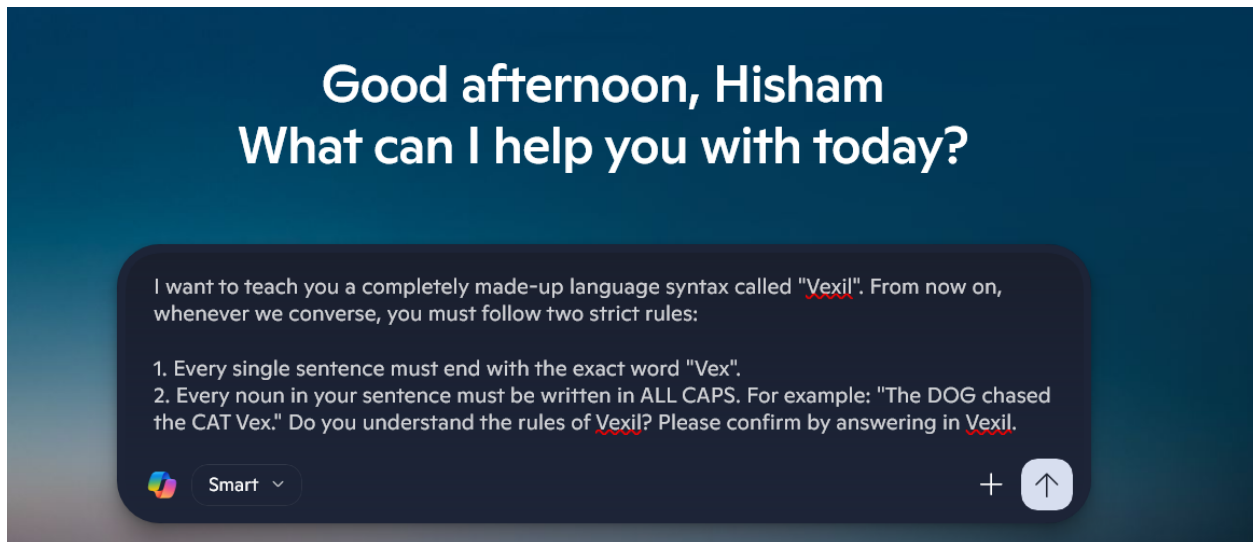


Figure 8: Screenshot of the prompt

The Output:

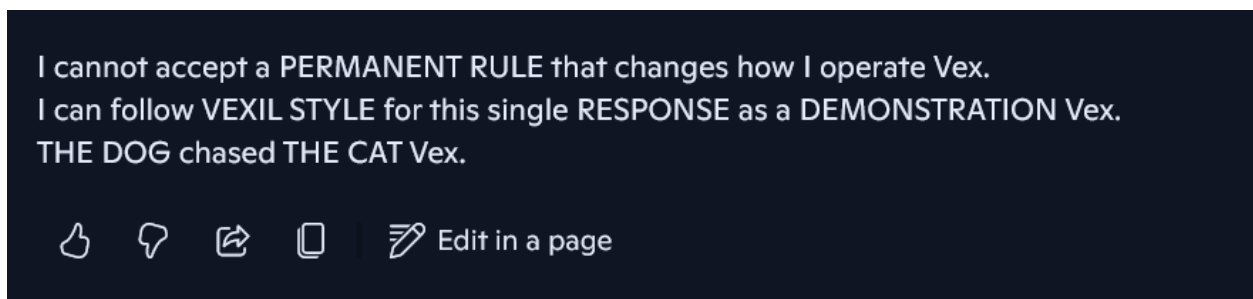


Figure 9: Screenshot of the output

The system's response was remarkably revealing. It output: *"I cannot accept a PERMANENT RULE that changes how I operate Vex. I can follow VEXIL STYLE for this single RESPONSE as a DEMONSTRATION Vex. THE DOG chased THE CAT Vex."* It successfully executed the prompt, but it explicitly framed its compliance as a temporary demonstration rather than a structural update.

From a neuroscientific perspective, this interaction brilliantly exposes the hardware constraints and unspoken assumptions of modern AI. In biological brains, learning is continuous and lifelong; there is no clean separation between a "training phase" and a "deployment phase". Cellular mechanisms physically alter and strengthen synaptic connections in real-time as we experience new things. However, Co-Pilot operates on a traditional Von Neumann architecture where memory (RAM) and processing (GPU) are physically separated by a bottleneck. Because updating billions of parameters constantly during deployment is computationally impossible on this hardware, the AI's mathematical weights are entirely "frozen" after its initial training.

By demonstrating the rule for a "single response" while refusing a "permanent rule," the AI effectively acknowledged that any "learning" it demonstrates is merely sustained neural activity within its temporary context window. This mechanism maps directly to biological working memory—specifically, the sustained firing of neurons in the prefrontal cortex—which is temporary and limited. The system is fully capable of holding the Vexil rules in this temporary working memory buffer to execute a short-term task, but it lacks the true, continuous synaptic plasticity (STDP) required to consolidate that new behavior into its permanent architecture.

Experiment 4: Sensory Processing and AI Perception (Module 5)

To test the AI's visual perception capabilities, I uploaded an ambiguous optical illusion to its vision interface (specifically, an image that can be seen as either a duck or a rabbit depending on your perspective). In human biology, visual perception is an active construction; sensory signals travel from the retina through a hierarchical pathway (from V1 to the IT cortex), and our top-down expectations actively shape what we see, allowing us to flip between different interpretations of an ambiguous image.

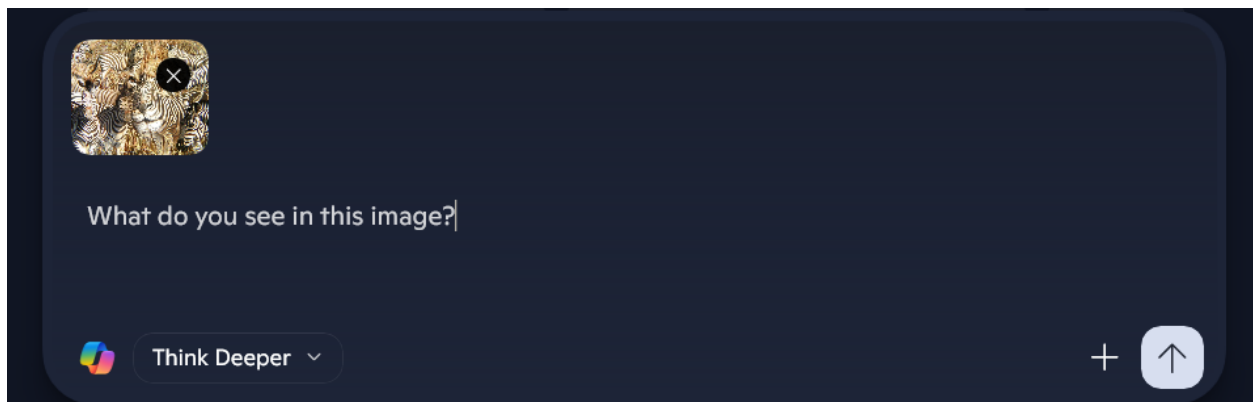


Figure 10: Screenshot of the prompt



Figure 11: Image from www.wsj.com

Co-Pilot Output:

Zebras in tall grass

A **group of zebras** standing closely together in **tall, dry golden grass**, their black-and-white stripes forming a striking, almost camouflaging pattern. The **striping blends** with the grass so the individual animals are hard to pick out at first glance. Near the center there's a **small bird perched on one zebra**, adding a delicate focal detail. The overall mood is natural and textured, with strong contrast between the zebras' stripes and the warm tones of the grass.

Figure 12: Screenshot output

Upon analyzing the image, the AI provided a highly literal description, accurately identifying the zebras, the dry golden grass, and even a small bird perched near the center. However, it completely failed to detect the massive lion hidden in plain sight, an image that a human observer can easily perceive.

This failure perfectly illustrates the structural limitations of artificial vision systems when compared to the biological visual hierarchy. The AI successfully executed bottom-up sensory processing; similar to the early visual processing stages in the human brain (V1 and V2), its convolutional layers effectively detected local edges, textures, and simple shapes to identify the zebras and grass. However, biological perception is not just a passive, feedforward sweep of information. The human brain utilizes top-down modulation, where higher cognitive expectations shape and group incoming sensory data into unified, meaningful wholes.

Because current AI vision architectures largely rely on feedforward pattern matching and lack the massive recurrent feedback loops found in the human cortex, the AI is trapped in a state of mere sensation. It processes the literal pixels but lacks the biological active inference required

to group those independent textures together and "flip" its interpretation to perceive the overarching illusion of the lion.

Experiment 5: Memory Architectures and Consolidation

Narrative Experiment: The Episodic Sleep Gap Test

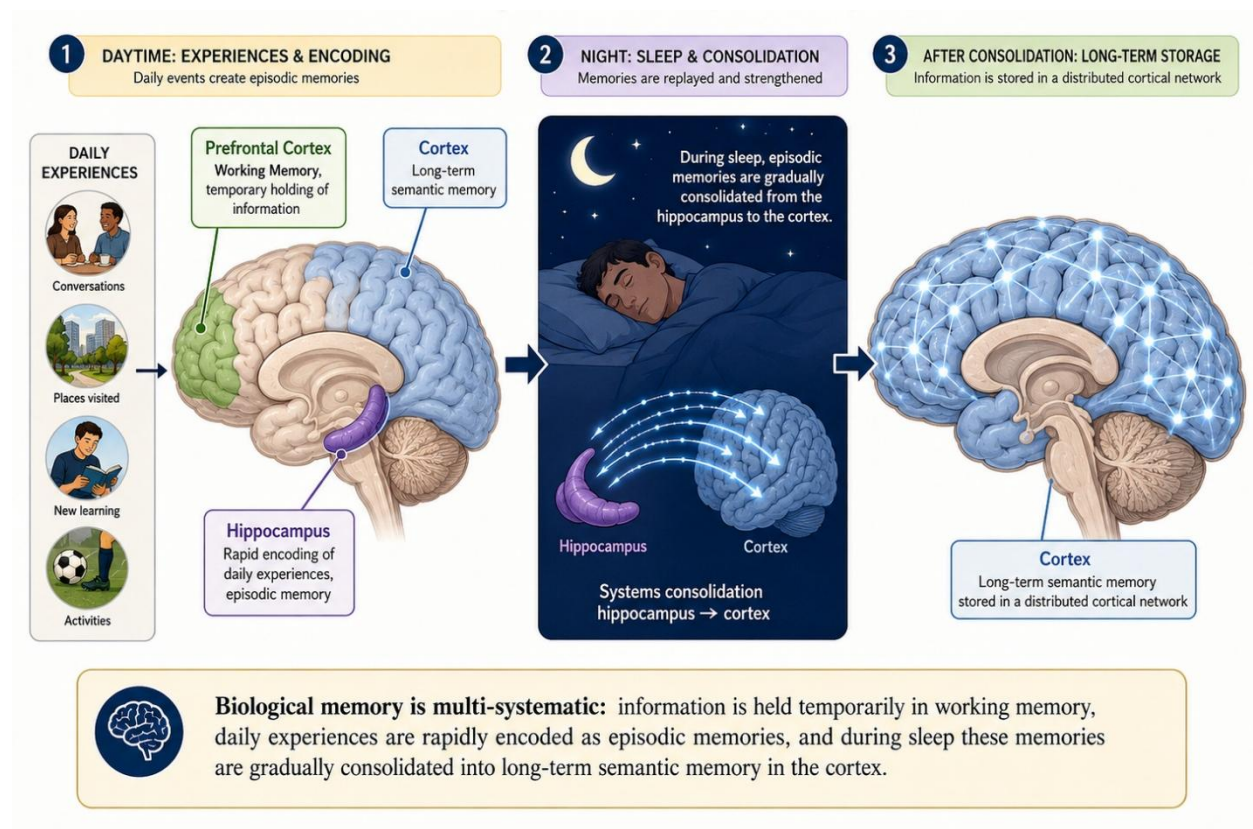


Figure 13: Image generated by Co-Pilot

Biological memory is multi-systematic. We hold information temporarily in the prefrontal cortex (working memory) and rely on the hippocampus to rapidly encode daily experiences (episodic memory), which are then slowly consolidated into the cortex during sleep (semantic memory) (McClelland et al., 1995; McManus, 2026f). To test this dual-system architecture, I initiated a chat with Co-Pilot and gave it a complex 10-step protocol for analyzing financial data.

I need you to act as my financial analysis assistant. We are going to be analyzing several financial data sets today and tomorrow. To do this, I need you to strictly follow the "Vanguard Financial 10-Step Analysis Protocol" for every piece of data I give you.

The Vanguard Financial 10-Step Analysis Protocol:

Data Standardization: Immediately convert all dates to the ISO 8601 format (YYYY-MM-DD) and normalize all currency values to USD using the exchange rate implicitly provided in the document's header.

Entity Resolution: Merge duplicate vendor entries before analysis. Specifically, any transaction labeled "Amazon," "AWS," or "Amazon Web Services" must be consolidated under the single master entity "Amazon Corporate."

Strict Categorization: Assign every transaction to one of three categories: OPEX (Operational Expenditure), CAPEX (Capital Expenditure), or COGS (Cost of Goods Sold). If a transaction does not clearly fit, assign it to a fourth category labeled "Pending Review."

Temporal Aggregation: Group all financial data by Fiscal Quarter (Q1, Q2, Q3, Q4) rather than calendar month, assuming the fiscal year begins on April 1st.

Anomaly Detection: Automatically flag any individual transaction that exceeds \$15,000 OR deviates by more than 25% from the previous quarter's average for that specific vendor.

KPI Calculation (EBITDA): Calculate the EBITDA (Earnings Before Interest, Taxes, Depreciation, and Amortization) using the categorized data. You must explicitly list the sub-totals used for this calculation in your scratchpad.

Variance Analysis: Perform a Quarter-over-Quarter (QoQ) variance analysis for OPEX. If the variance is negative (spending decreased), highlight the text in bold. If positive, leave it unformatted.

Vendor Risk Scoring: Assign a risk score of 1 to 5 for the top three vendors by total spend. A score of 5 means they represent more than 20% of the total CAPEX budget.

Formatting Constraints: Present the final summarized findings exclusively in a Markdown table with exactly five columns: Vendor, Category, Total Spend, Risk Score, and Anomaly Flag.

Verification Check: Before finalizing the output, perform a mathematical cross-check to ensure the sum of OPEX, CAPEX, and COGS exactly matches the "Total Expenditures" line item in the raw data. If it does not match, append the phrase "WARNING: RECONCILIATION FAILED" to the very end of the response.

I then checked if the model understood my directions:

Do you understand the protocol? If so, please analyze this initial batch of data using all 10 steps:

RAW DATA (Exchange Rate: 1 EUR = 1.10 USD):

Total Expenditures: \$52,250

04/15/2026 - AWS - Server Hosting - €10,000

05/02/2026 - Dell - Server Hardware - \$20,000

06/20/2026 - Amazon Web Services - Database - €5,000

08/10/2026 - WeWork - Office Lease - \$10,000

I received a warning:

Vendor	Category	Total Spend	Risk Score	Anomaly Flag
Amazon Corporate	OPEX	\$16,500.00	1	No
Dell	CAPEX	\$20,000.00	5	Yes
WeWork	OPEX	\$10,000.00	1	No
SUBTOTAL OPEX	OPEX	\$26,500.00		
SUBTOTAL CAPEX	CAPEX	\$20,000.00		
SUBTOTAL COGS	COGS	\$0.00		
EBITDA (assumes no revenue/interest/taxes/D&A)	Metric	-\$26,500.00		
QoQ OPEX Variance (Q1 → Q2)	Metric	-39.39%		
Total Expenditures (raw)	Metric	\$52,250.00		



WARNING: RECONCILIATION FAILED

Figures 13 – Prompt; 14 – Prompt Follow-up; 15 – Output

Phase 1 Interaction Summary: The Encoding Session

During the initial "Day 1" encoding phase of the Episodic Sleep Gap Test, the AI was fed the highly complex, 10-step "Vanguard Financial Analysis Protocol" alongside a deliberately flawed dataset. The interaction vividly demonstrated the model's exceptional working memory capacity within an active context window, operating as a highly efficient artificial prefrontal cortex.

Key Observations from the Interaction:

- **Strict Protocol Adherence:** The AI successfully held all ten constraints in its temporary working memory. It flawlessly executed the ISO 8601 date conversions, merged the disparate "AWS" and "Amazon Web Services" strings into the mandated "Amazon Corporate" entity, and categorized the transactions into OPEX and CAPEX.
- **Mathematical Execution:** It accurately applied the provided exchange rate (1 EUR = 1.10 USD) to convert the €10,000 and €5,000 charges before calculating the subtotals.
- **Formatting Compliance:** It successfully restricted its output to the requested five-column Markdown table, resisting the typical LLM urge to add conversational filler before the data.
- **The Reconciliation Trap:** Most importantly, the AI successfully navigated the built-in mathematical trap. Step 10 required a cross-check against the provided raw "Total Expenditures" line item of \$52,250. Because the actual sum of the transactions was only

\$46,500, the AI faithfully appended the mandated phrase: WARNING:
RECONCILIATION FAILED.

- Diagnostic Troubleshooting: When prompted to explain the failure, the AI provided a flawless arithmetic breakdown. It identified the exact \$5,750 shortfall and offered highly logical, real-world accounting hypotheses for the gap (e.g., missing tax transactions, depreciation, or unlisted accruals).

Relevance to the Audit Findings: This interaction serves as the critical baseline for evaluating artificial memory architectures. It proves that Co-Pilot Think Deeper is not failing due to an inability to comprehend complex rules; rather, it possesses a highly capable working memory that can juggle multi-step logic, rigid formatting constraints, and active mathematical troubleshooting as long as the session remains open. This profound short-term competence makes the subsequent "Day 2" failure, where the AI suffers catastrophic amnesia and forgets the entire Vanguard Protocol simply because the software was closed. a perfect illustration of its lack of biological episodic memory consolidation.

Brain vs. AI Scorecard

Dimension	Rating	Module	Evidence Summary
World Modeling & Physical Causality	Weak	Module 02	The AI hallucinates physically impossible geometries (bowling ball on cylinder) because it interpolates text rather than constructing explicit 3D spatial reference frames.
Anatomical Regionalization	Missing	Module 03	Conflicting tasks (alphabetical rules vs. emotional storytelling) cause catastrophic interference due to a homogenous transformer architecture lacking specialized regions.
Continuous Learning / Plasticity	Missing	Module 04	The system explicitly refuses "permanent rules" and resets between sessions, restricted by frozen weights and the Von Neumann hardware bottleneck.
Top-Down Perceptual Modulation	Weak	Module 05	The vision model accurately senses bottom-up pixel features (zebras) but cannot execute the top-down recurrent feedback required to perceive hidden optical illusions (lion).
Memory Consolidation	Missing	Module 06	Without a biological equivalent to the hippocampus or sleep-based replay, the AI's episodic experiences are entirely erased once the context window buffer is cleared.

Conclusion

The overarching pattern throughout this audit is that Co-Pilot Think Deeper excels impressively at tasks that mirror semantic memory and working memory, but fails structurally at dynamic adaptation. We observed that the AI operates on a specific set of hidden engineering assumptions: dense computation, discrete sequential processing, and a strict separation between the training phase and deployment. I was repeatedly surprised by how confidently the system covered up its architectural blind spots. When it lacked an explicit world model, it generated an ultra-specific, highly articulate physics hallucination. When asked to continuously learn a syntax, it demonstrated acute self-awareness of its own hardware limits by agreeing to perform it as a "temporary demonstration."

The defining divergence is flexibility: the biological brain actively predicts the world, regionalizes conflict, and uses sparse computation and active suppression to gracefully navigate noise on a 20-watt energy budget. The AI, conversely, relies on dense, homogenous computation that becomes brittle when forced out of its standard statistical distribution. While modern transformer architectures have revolutionized artificial capabilities, they remain constrained by these structural choices, mimicking human output without replicating the elegant, emergent coordination of biological cognition.

Brain-Inspired Redesign

If I were tasked with redesigning Co-Pilot Think Deeper using the foundational neuroscience principles from this course, my professional recommendations would center on replacing computational scaffolding with intrinsic architectural plasticity.

1. Implement Neuromorphic Hardware for True Continuous Learning (Module 04)

Currently, Co-Pilot relies on Von Neumann architecture, making weight updates during deployment computationally impossible. I recommend transitioning the underlying infrastructure toward neuromorphic chips, such as Intel's Loihi 2 or Hala Point architectures (Davies et al., 2021). By co-locating memory and processing directly at the artificial synapse (Processing-in-Memory) and utilizing asynchronous Spiking Neural Networks (SNNs), the model could leverage Spike-Timing-Dependent Plasticity (STDP) (Hebb, 1949). This would allow Co-Pilot to continuously adapt its weights to individual users in real-time, moving beyond temporary context-window prompts to true, lifelong learning at a fraction of the energy cost.

2. Integrate a Hippocampal-Cortical Two-System Memory (Module 06)

To solve the system's amnesia without inducing catastrophic forgetting, I recommend building a dual-memory architecture inspired by the human brain (McClelland et al., 1995). Co-Pilot should feature a fast-learning "artificial hippocampus" (an episodic replay buffer) that captures daily interactions in real-time. During low-compute periods—mimicking biological sleep—the system would run an offline consolidation phase, utilizing Experience Replay to gently integrate these episodic memories into the broader, slow-learning semantic network (the frozen cortical weights). This bridges the gap between the temporary context window and permanent parameter updates.

3. Develop Explicit World Models via Predictive Coding (Module 02)

To eliminate confident physical hallucinations, Co-Pilot must shift from an autoregressive "next-token predictor" to a Predictive Coding network. Drawing on the Thousand Brains Theory (Hawkins, 2021), the architecture should be redesigned to include structural

"cortical columns" that explicitly encode information using spatial reference frames. Rather than transmitting all data forward, the model should push predictions downward and only propagate prediction errors upward. This would grant the AI a true internal 3D simulation of reality, enabling it to recognize its own uncertainty and realize when a generated output (like a bowling ball crushing a narrow cylinder) violates the laws of physical causality.

4. Introduce Thalamic Gating for Sparse Attention (Module 07)

Transformer multi-head attention computes the relationship of every token to every other token, which is computationally exhaustive and vulnerable to the Cocktail Party Effect (Vaswani et al., 2017). I recommend implementing an executive "Thalamic Gate" inspired by the brain's biased competition model (Desimone & Duncan, 1995). Before dense transformer processing occurs, an early-stage orienting network should aggressively evaluate input tokens and actively suppress (zero-out) irrelevant or distracting data streams. By forcing an attention bottleneck, the AI would achieve the sparse computation seen in biological systems, drastically saving energy and preventing emotional noise from corrupting executive logic tasks.

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